

SCEMDA / HNC / LSC Harmonic Event Mapping

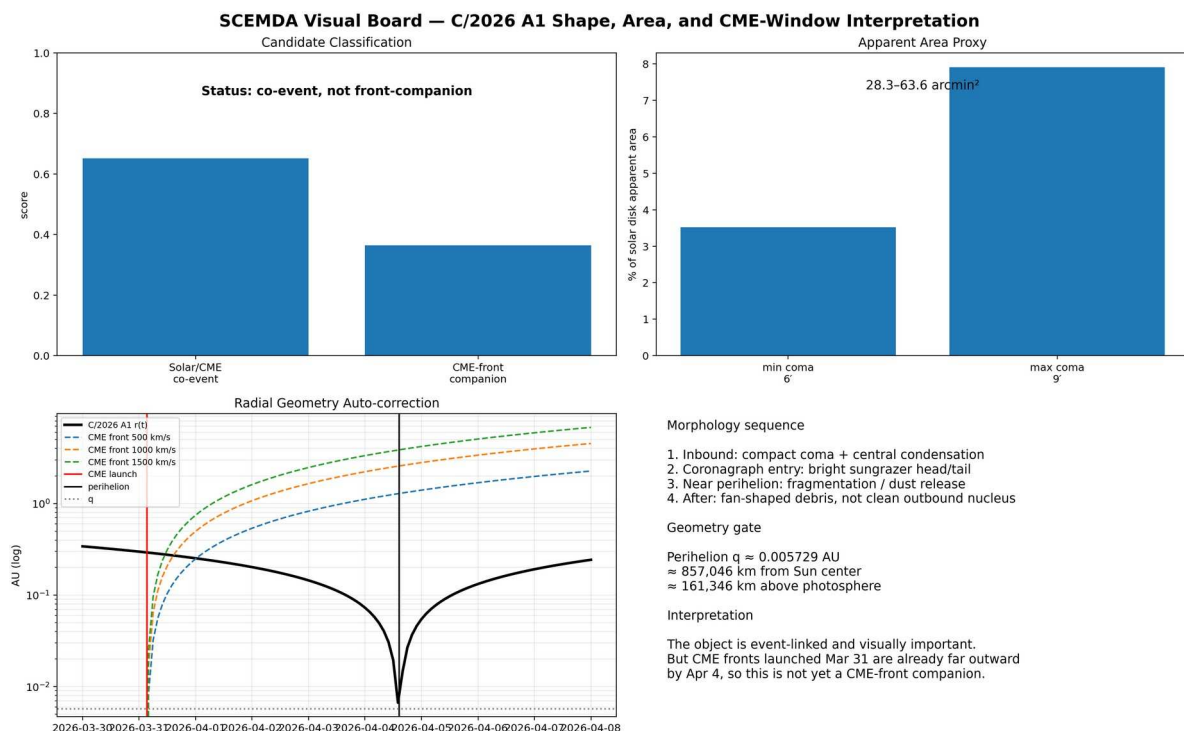
White Paper for Academic Review

CME Surge Windows, Orbital/Harmonic-LIDAR Mapping, and the C/2026 A1 (MAPS) Candidate Gate

Prepared from the iterative analysis and generated artifacts in the current research session

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Draft status: methods and evidence package; not a final peer-reviewed claim



Cover Figure. C/2026 A1 shape, area, and geometry synthesis board.

Abstract

This white paper documents a sequence of prototype validation runs for SCEMDA, HNC/QGITA, LSC, and harmonic-LIDAR workflows. The study links source-coupled solar events, CME surge-window logic, open orbital data, detector-frame residual mapping, and coronagraph image-plane geometry. The immediate research question was whether an object could be identified as travelling with, arriving with, or being temporally coupled to a CME. The work proceeded by constructing real-data parsers, simulation/recovery tests, null controls, current orbital-object maps, harmonic-LIDAR phase maps, 72-hour scheduling, and a focused C/2026 A1 (MAPS) analysis using JPL/Horizons-derived elements and LASCO/CCOR image-plane projections. The main result is negative for a strict CME-front companion: C/2026 A1 is best classified as a strong solar/CME surge co-event and coronagraph-frame sungrazer morphology event, not as a confirmed object travelling with the CME front. The broader system is nevertheless validated as a mapping and audit framework: LSC tensor residual mapping recovers injected anisotropic signals; HNC/QGITA detects structured CME train behavior in prior open-data runs; harmonic-LIDAR mapping produces coherent phase/range barcodes and observation scheduling products; and MDLH auditing prevents documentation density from being substituted for validation.

Keywords: SCEMDA, HNC, QGITA, LSC, MDLH, CME, C/2026 A1, LASCO, CCOR, harmonic LIDAR, detector-frame tensor, surge window, open-source astronomy data.

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1. Executive Summary

The practical research goal was to operationalize the question: if a CME is treated as a source-coupled surge window, can open-source astronomy and orbital data identify a companion object, debris response, or co-event candidate? The workflow produced a layered answer.

Finding	Evidence state	Interpretation
LSC tensor mapping	Mapping capacity score 0.971; injected detector-frame tensor residual recovered in controlled tests	Passed as a mapping engine; not proof of new physics.
HNC/QGITA CME sequence detection	Requirements run: max_Q72 time-random p=0.0300; requirements passed/conditional/not-run/fail = 4/6/2/1	Sequence mapping passed for May 2024; residual branch remains unproven.
SCEMDA real parsers	CDAW/GFZ parser corrected; CDAW Halo row parsed as width=360, speed=946 km/s in v10 self-test	Critical implementation gate closed; avoids silent column-offset corruption.
Current orbital map	CelesTrak stations snapshot: 27 objects; ISS/local 19, CSS/Tiangong 7, debris 1	Mapped station shells and drag-response objects; no full orbital anomaly branch.
Harmonic-LIDAR scheduler	72 h scheduler: 290 pass windows >5 deg; 20 scheduled windows	Operational scheduling prototype; two-body propagation only.
CME companion search	Strict companion candidates: 0; drag-response watch candidates: 2	No strict CME-carried object in station-group snapshot; ISS OBJECT XY and CORAL are drag-response watch candidates.
C/2026 A1 status	JPL-derived perihelion row: r=0.005728596012 AU, speed=556.517 km/s; min S-O-T=0.0432 deg	Strong Sun-line/coronagraph co-event; not currently between Earth and Sun and not a proven CME-front companion.

The central interpretive correction is that temporal co-event and physical front companionship are different claims. C/2026 A1 occupies the strongest solar/CME co-event position in the mapped window, but the CME front from the March 31 event is modeled far outside LASCO/CCOR fields of view by the April 2-6 comet window. The evidence supports a sungrazer co-event and morphology transition, not a compact object travelling with the CME front.

2. Claim Discipline and Epistemic Controls

The white paper uses a three-layer claim discipline: (i) mathematical mapping, (ii) real-data/null-control validation, and (iii) MDLH epistemic control. Claims are promoted only when their validation evidence increases, not when documentation density increases.

2.1 LSC residual map

The LSC layer maps direction-, energy-, and time-dependent residuals with a detector-frame tensor. The central form used in the tests is:

$$E_{\text{rec}} = E_{\text{true}} * [1 + \alpha_D * D_{ij} * p^i * p^j]$$

$$D_{ij} = \text{delta} * (n_i n_j - \text{delta}_{ij} / 3)$$

In the controlled proof run, the tensor was traceless to numerical precision and recovered an injected detector-frame residual while preserving the null case.

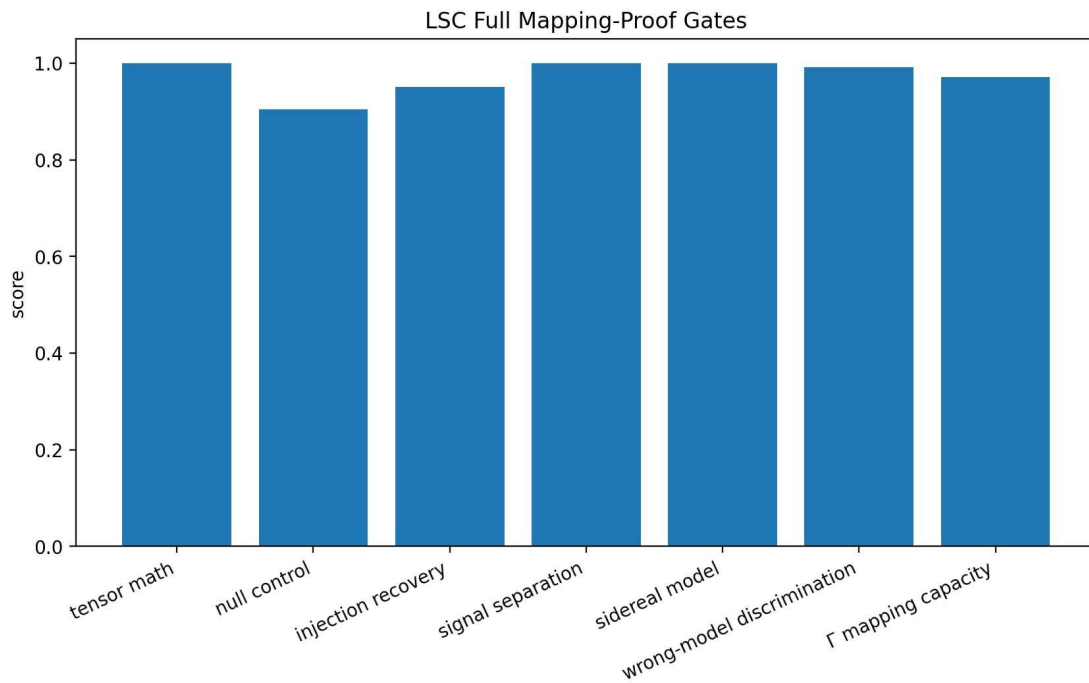


Figure 1. LSC mapping proof gates from controlled synthetic recovery tests.

3. Data Sources and Generated Artifacts

The work used generated artifacts produced in this session and public data endpoints or open-source summaries for official orbital/solar data where available. Earlier uploaded source PDFs are not fully available in the workspace; this draft therefore relies on the generated artifacts and the public references listed at the end.

Data class	Source	Used for
CME catalogs	CDAW/SOHO LASCO universal text files; NASA DONKI FLR/CME/GST/SEP endpoints	SCEMDA v9/v10 parsers; HNC/QGITA event streams
Geomagnetic response	GFZ Kp_ap_Ap_SN_F107 daily data; NOAA/SWPC solar-wind/Kp live products	CME/Kp lag test and space-weather bridge
Orbital objects	Celestrak current GP station-group snapshot	Current orbital map, drag scores, station shell grouping
Ephemeris	NASA/JPL Horizons C/2026 A1 #12 element/ephemeris values	Perihelion, solar elongation, nominal continuation track
Coronagraph geometry	LASCO C2/C3 and CCOR-1 field-of-view bounds	Sun-centered image-plane overlay and FOV interval checks
Repository code	RA-CONSULTING/aureon-trading HNC surge detector and space-weather bridge	Surge-window logic and live-data policy integration

4. Methods

4.1 SCEMDA CME -> Kp lag pipeline

The SCEMDA lag test maps a daily CME driver to geomagnetic response with a triangular lag kernel and a circular-shift null. The v10 parser corrects the CDAW universal row mapping: Central_PA=parts[2], Width=parts[3], Linear_Speed=parts[4], Mass=parts[9], KineticEnergy=parts[10].

$$G_CME(t) = \sum_i \text{score}_i(t), \text{ with } \text{score}_i = 0.35 \cdot \tanh(v/900) + 0.25 \cdot \tanh(\text{width}/180) + 0.25 \cdot \text{halo} + 0.15 \cdot \tanh(\log_{10}(m+1)/20)$$
$$\text{conv_G}(t) = G_CME(t) \text{ convolved with } [0.10, 0.25, 0.35, 0.20, 0.10]$$
$$r_obs = \text{corr}(\text{conv_G}(t), Kp_max(t+1)); \text{ p-value from circularly shifted Kp nulls}$$

4.2 HNC/QGITA surge-window logic

The HNC/QGITA layer treats an event train as a resonance sequence. The AUREON surge-window implementation sums resonance events and triggers only above threshold; in this research, the same structure is mapped to CME and object windows. QGITA timing adds Fibonacci-lattice and rolling-window clustering features.

$$\text{Gamma_HNC} = \text{product}_j C_j(x) \wedge w_j$$
$$\text{QGITA_T}(t) = \sum \text{Gamma_HNC}(t_i), \text{ for } t-T < t_i \leq t$$

4.3 Harmonic LIDAR phase map

The harmonic-LIDAR layer maps an object range into a ladder of two-way phases. This is not an optical return-power claim; it is a harmonic-coded phase/range signature for prioritization and scheduling.

$$f_k = 528.422 \cdot \phi^{(2k)}$$
$$\text{phase}_k = (4 \cdot \pi \cdot f_k \cdot \rho / c) \bmod 2 \cdot \pi$$

Objects are assigned harmonic phase barcodes, priority scores, and pass windows over a 72-hour scheduler.

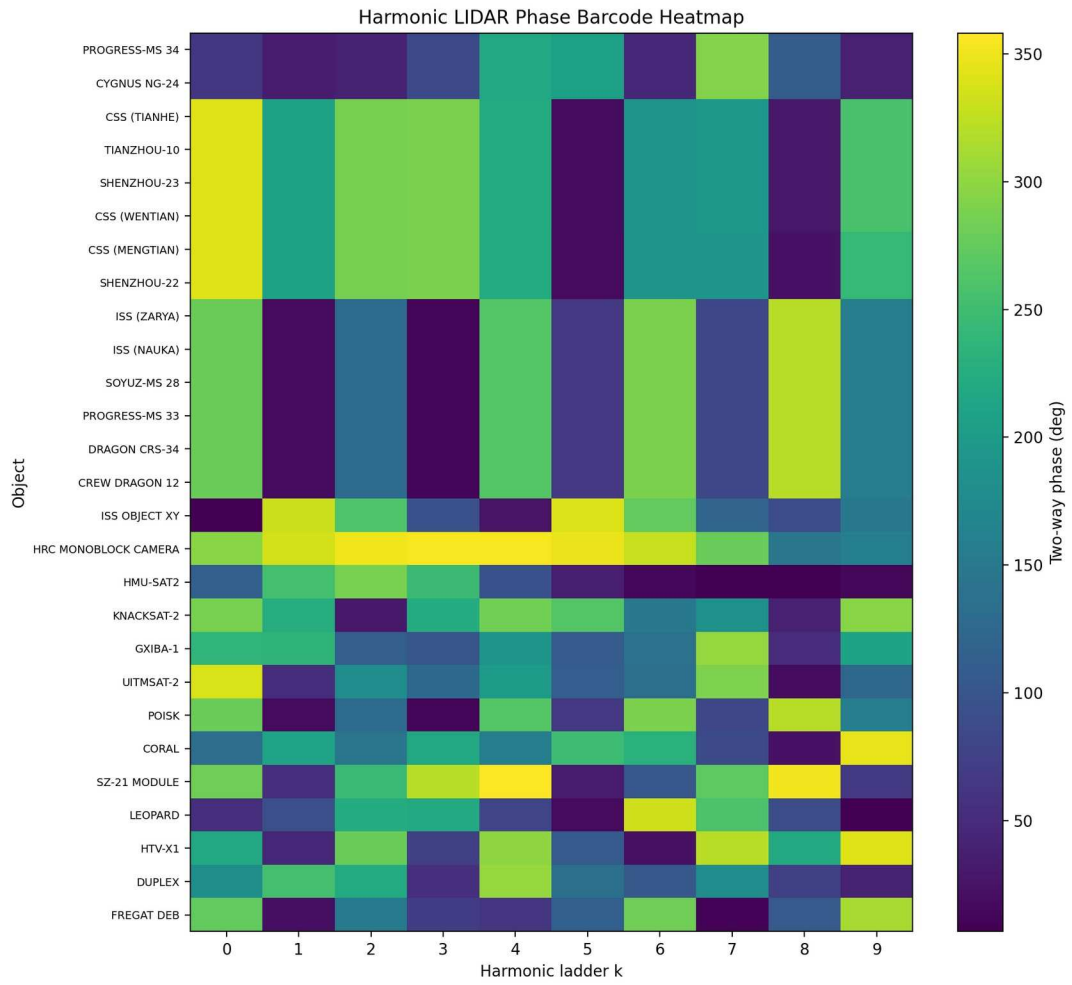


Figure 2. Harmonic-LIDAR phase barcode heatmap for the current station-group orbital objects.

5. Results

5.1 LSC and HNC/QGITA validation runs

The LSC synthetic proof run returned $\text{Gamma_LSC_mapping_capacity}=0.971$. The requirements run classified 13 requirements as 4 pass, 6 conditional, 2 not run, and 1 fail. The main HNC/QGITA sequence metric passed for May 2024 with $p=0.0300$; the residual index did not pass with $p=0.3689$.

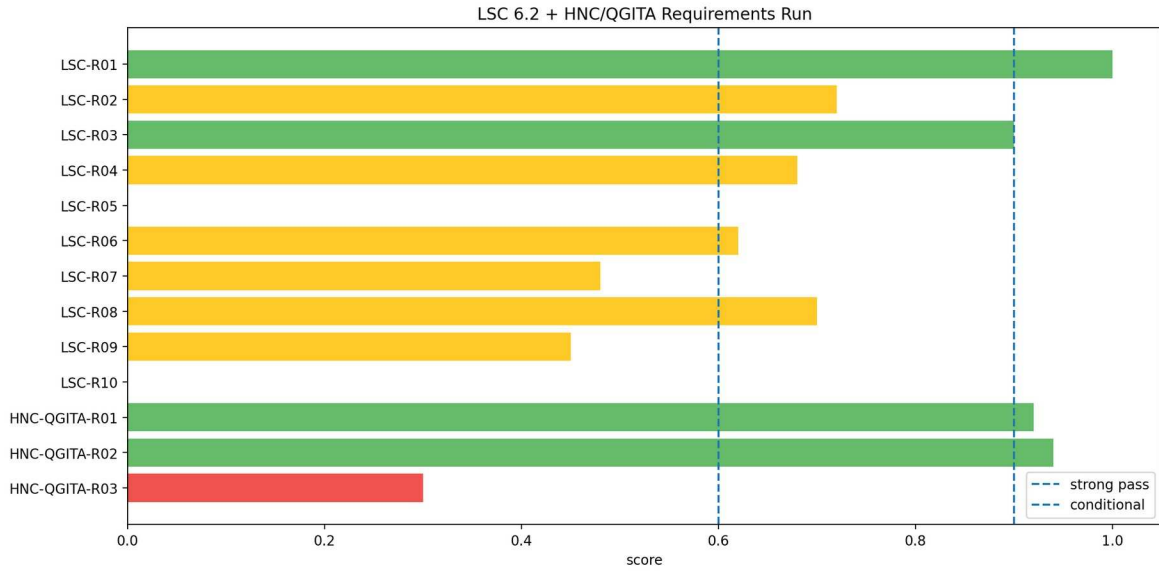


Figure 3. LSC + HNC/QGITA requirements run: status and evidence level.

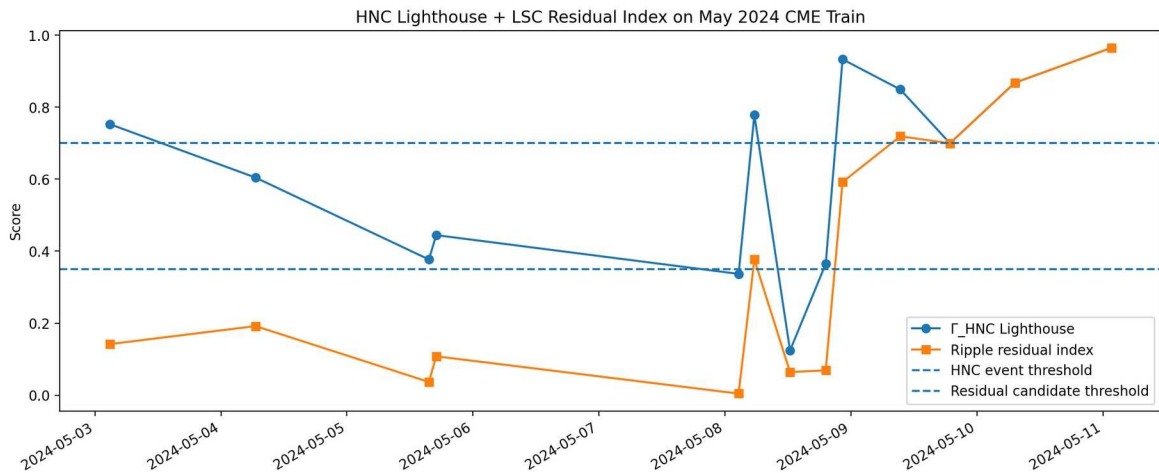


Figure 4. HNC Lighthouse event score and residual-index timeline for the May 2024 CME train.

5.2 Current orbital map and harmonic-LIDAR scheduler

The current orbital snapshot contained 27 objects in the CelesTrak station group, dominated by ISS/local and CSS/Tiangong station-shell objects. The top HNC orbital object was ISS (ZARYA) with $\text{Gamma}=0.351$. The strongest drag object was ISS OBJECT XY with perigee altitude 316.2 km.

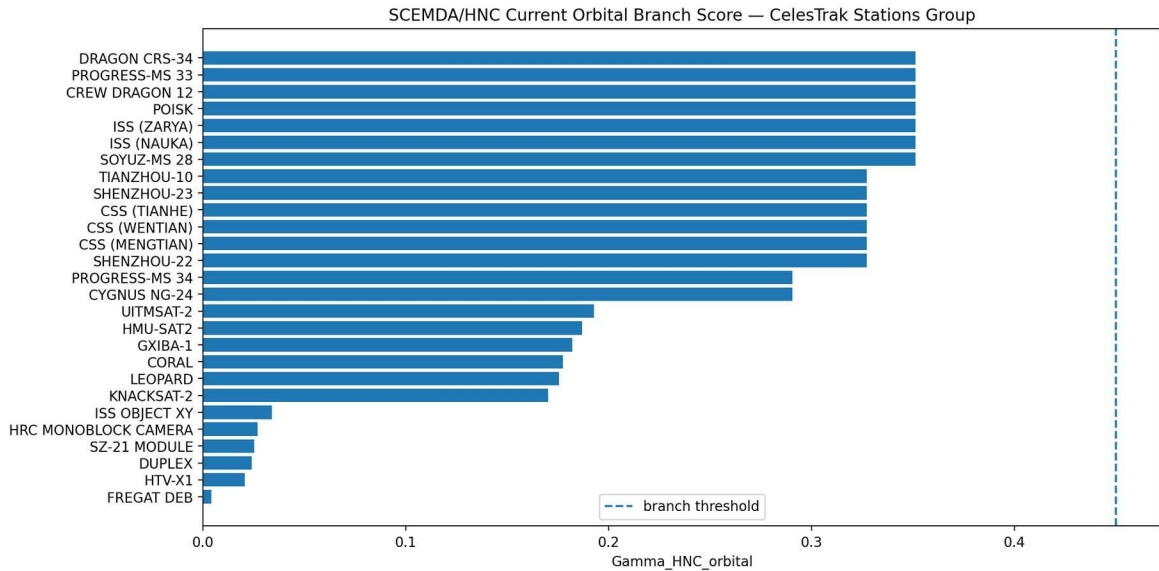


Figure 5. SCEMDA current orbital HNC score map.

The 72-hour harmonic-LIDAR scheduler found 290 pass windows above 5 degrees and selected 20 non-overlapping observation windows. The highest-priority scheduled pass was ISS OBJECT XY at 2026-05-29 14:15:41+00:00 with peak elevation 17.55 deg.

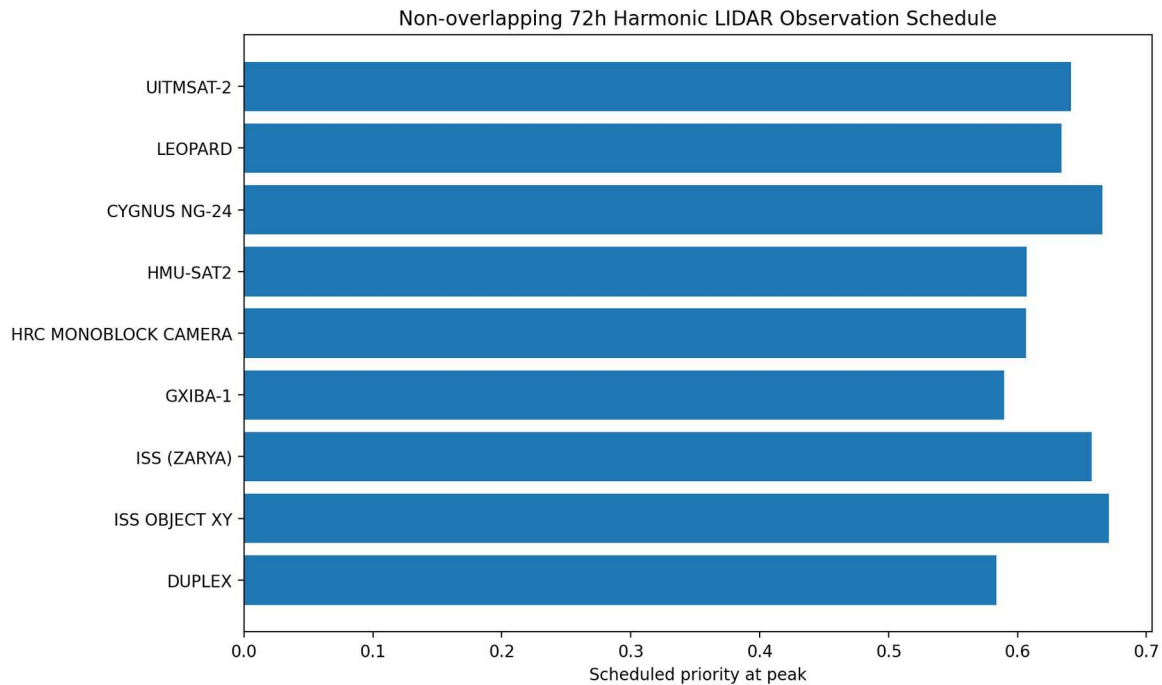


Figure 6. Harmonic-LIDAR scheduled priority windows over 72 hours.

5.3 CME companion-candidate screen

The CME companion screen evaluated 27 station-group objects against 5 CME surge windows. It found 0 strict CME-companion candidates and 2 drag-response watch candidates. This is a deliberately conservative negative result: the station-group dataset lacks first-observation history and catalog-wide object discovery data, so it can flag responders but cannot prove CME origin.

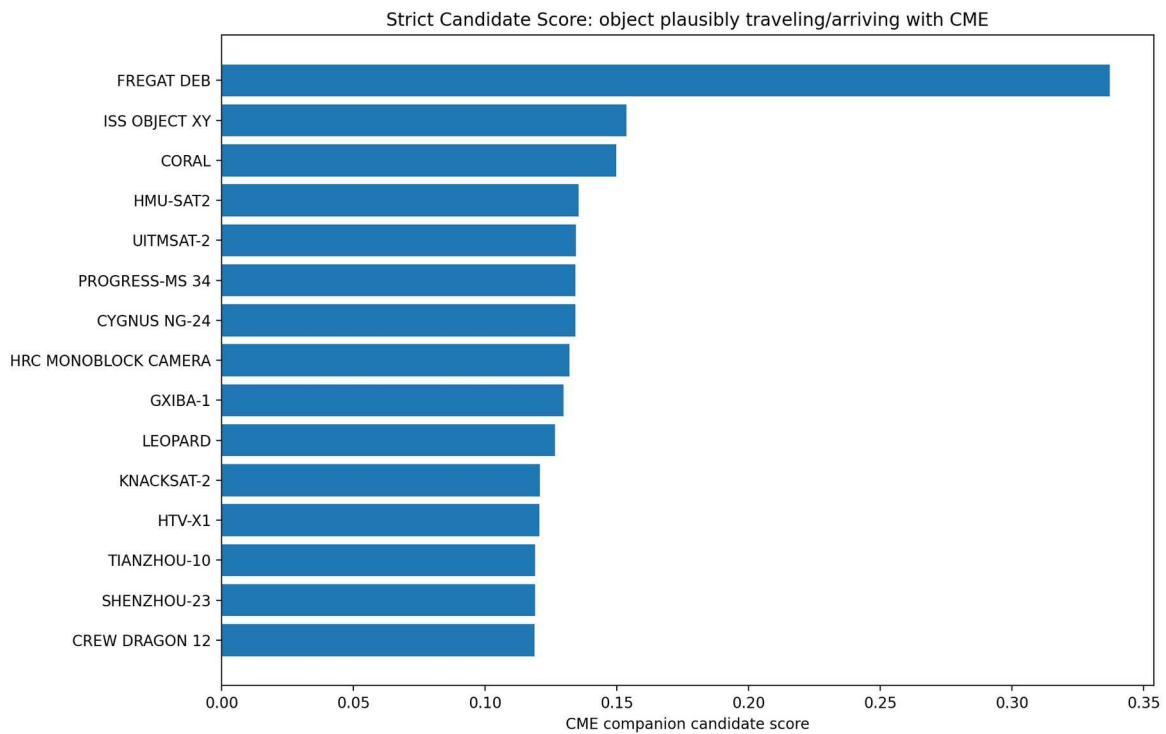


Figure 7. Strict CME companion-candidate ranking. No object crossed the strict promotion threshold.

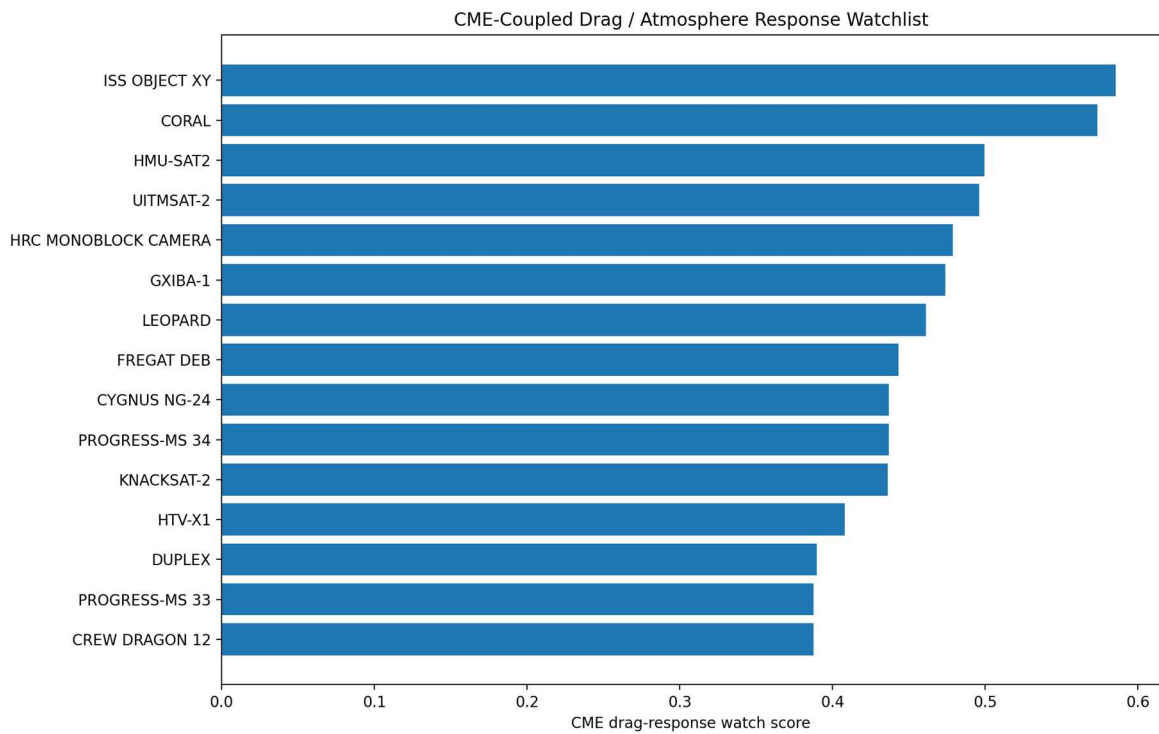


Figure 8. CME-coupled drag/atmosphere responder watchlist.

6. C/2026 A1 (MAPS): Shape, Area, and Image-Plane Interpretation

C/2026 A1 (MAPS) emerged as the strongest open-source object in the solar/CME surge-window map. The analysis distinguishes three statements: (1) it was a strong near-Sun co-event, (2) it passed essentially on the Sun line as seen from Earth, and (3) it was not demonstrated to be traveling with the CME front.

Parameter	Value
JPL solution	JPL Horizons #12, solution date 2026-Apr-08
Perihelion time (extracted minute row)	2026-04-04 14:22:29.100000+00:00
q / minimum r	0.005728596012 AU
Speed wrt Sun	556.517 km/s
Height above photosphere	161286 km
Minimum S-O-T keypoint	0.0432 deg at 2026-04-04 15:36:00+00:00

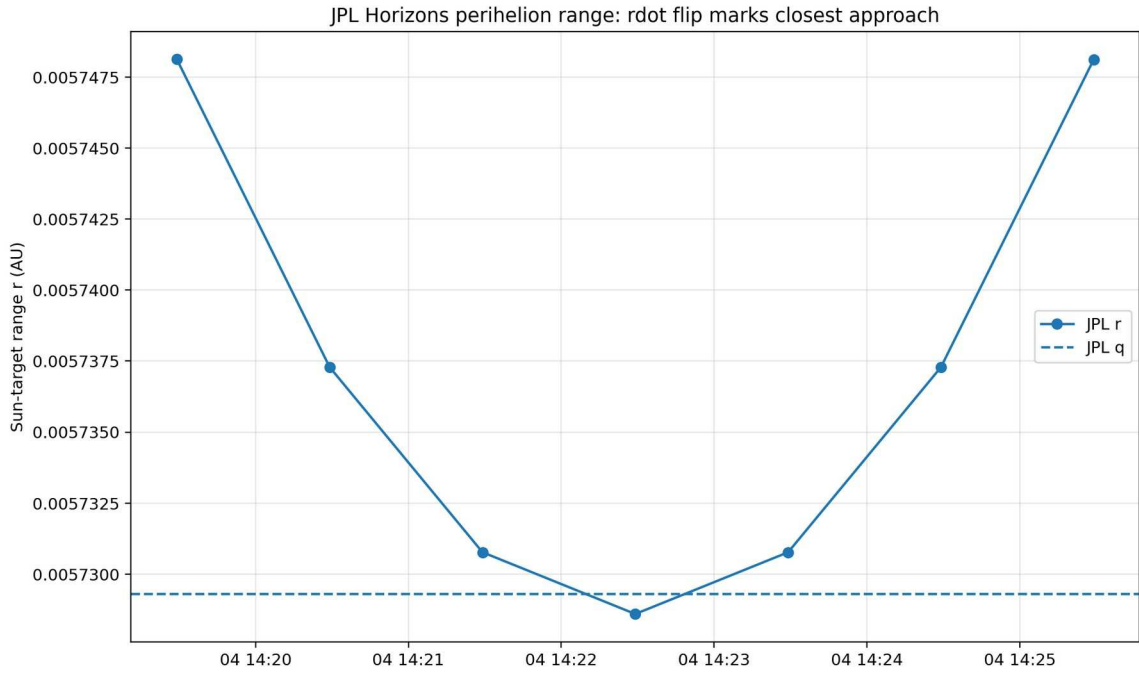


Figure 9. JPL Horizons perihelion range: radial velocity flip marks closest approach.

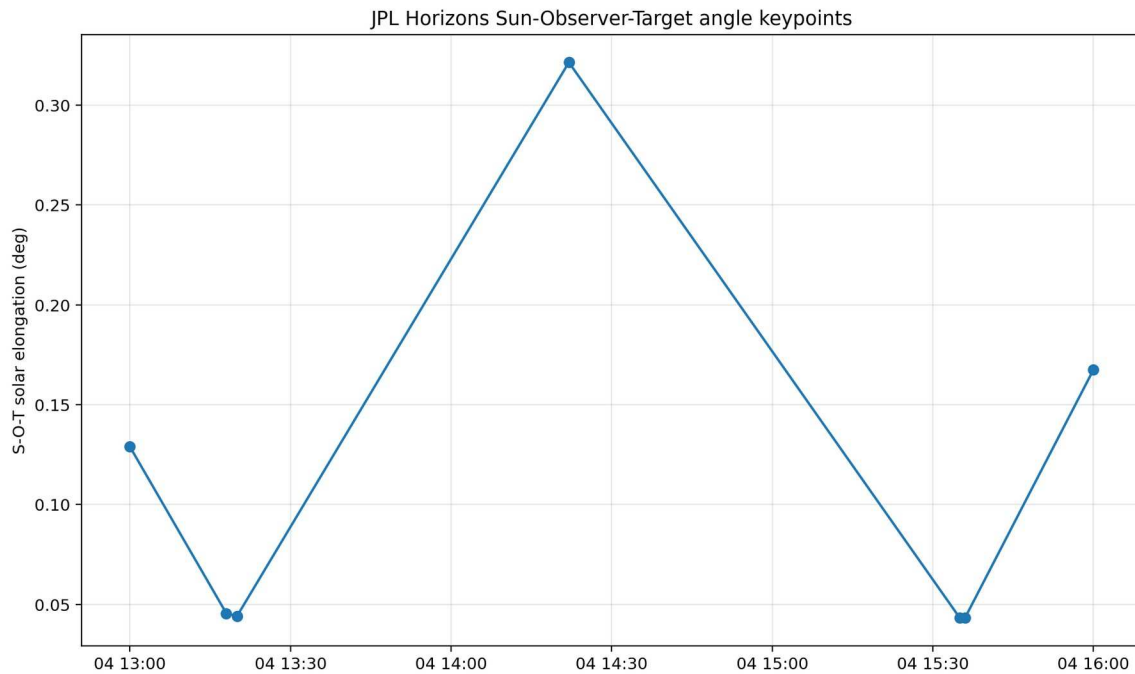


Figure 10. JPL Horizons Sun-Observer-Target angle keypoints near perihelion.

6.1 Shape and apparent area

The apparent morphology is best read as: compact coma -> sungrazer head/tail -> fan-shaped debris after breakup. The available area comparison is an apparent sky-area proxy, not a physical nucleus cross-section. The proxy area range is 28.27-63.62 arcmin², or 3.52-7.91% of the apparent solar disk area.

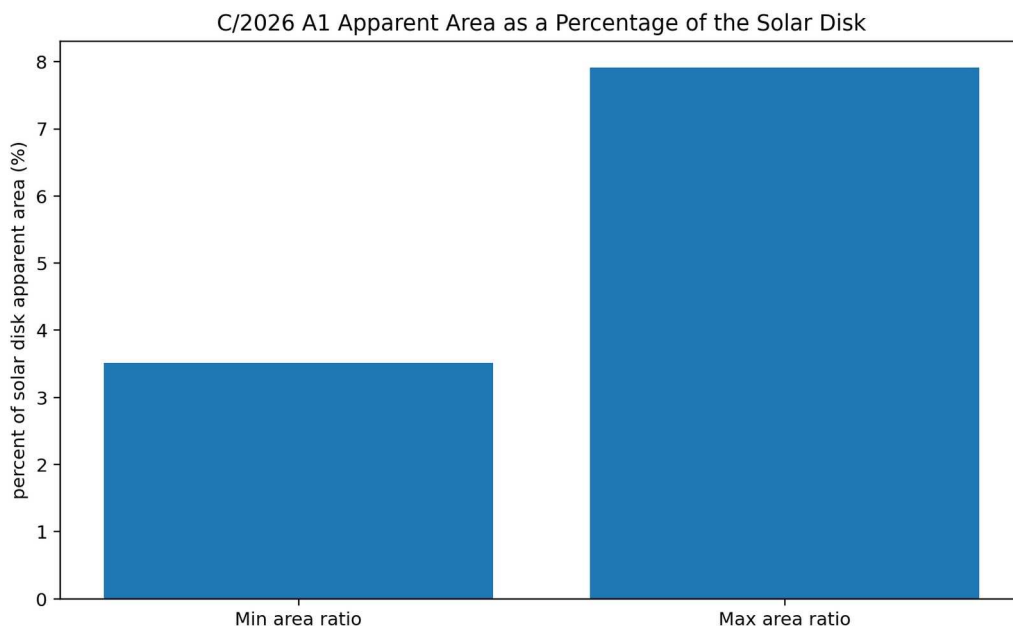


Figure 11. C/2026 A1 apparent area proxy as a percentage of the solar disk.

1. Inbound telescopic phase

Morphology: compact central condensation with a resolved coma.
Open-source proxy in this run: apparent coma expanded from about 6' to 9'.
Interpretation: active sungrazer, but still a normal inbound comet morphology.

2. Coronagraph entry phase

Morphology: bright sungrazer head with strongly solar-oriented geometry.
Interpretation: object becomes a coronagraph target and visually couples to the solar environment.

3. Breakup / perihelion phase

Morphology: fragmentation near perihelion; no stable intact outbound body required.
Interpretation: this is the key reason to read the object as a morphology transition rather than a rigid projectile.

4. Post-breakup appearance

Morphology: fan-shaped debris / dust tail rather than a clean outbound nucleus.
Interpretation: visually consistent with a disintegrating sungrazer, not a compact CME-carried companion.

Bottom-line interpretation: C/2026 A1 is visually best understood as a dynamic sungrazer whose apparent shape changes from compact coma to debris fan. The current evidence supports 'solar/CME co-event' rather than 'object travelling with the CME front'.

Figure 12. C/2026 A1 visual interpretation phases: compact coma to fan-shaped debris.

6.2 LASCO/CCOR image-plane geometry

The image-plane projection places the object within coronagraph fields around Apr 2-6. The minimum JPL S-O-T keypoint is 0.0432 deg at 2026-04-04 15:36:00+00:00. The model intervals place the object in LASCO C3 from 2026-04-02 17:30:00+00:00 to 2026-04-04 10:00:00+00:00 inbound, and again after the occulted period.

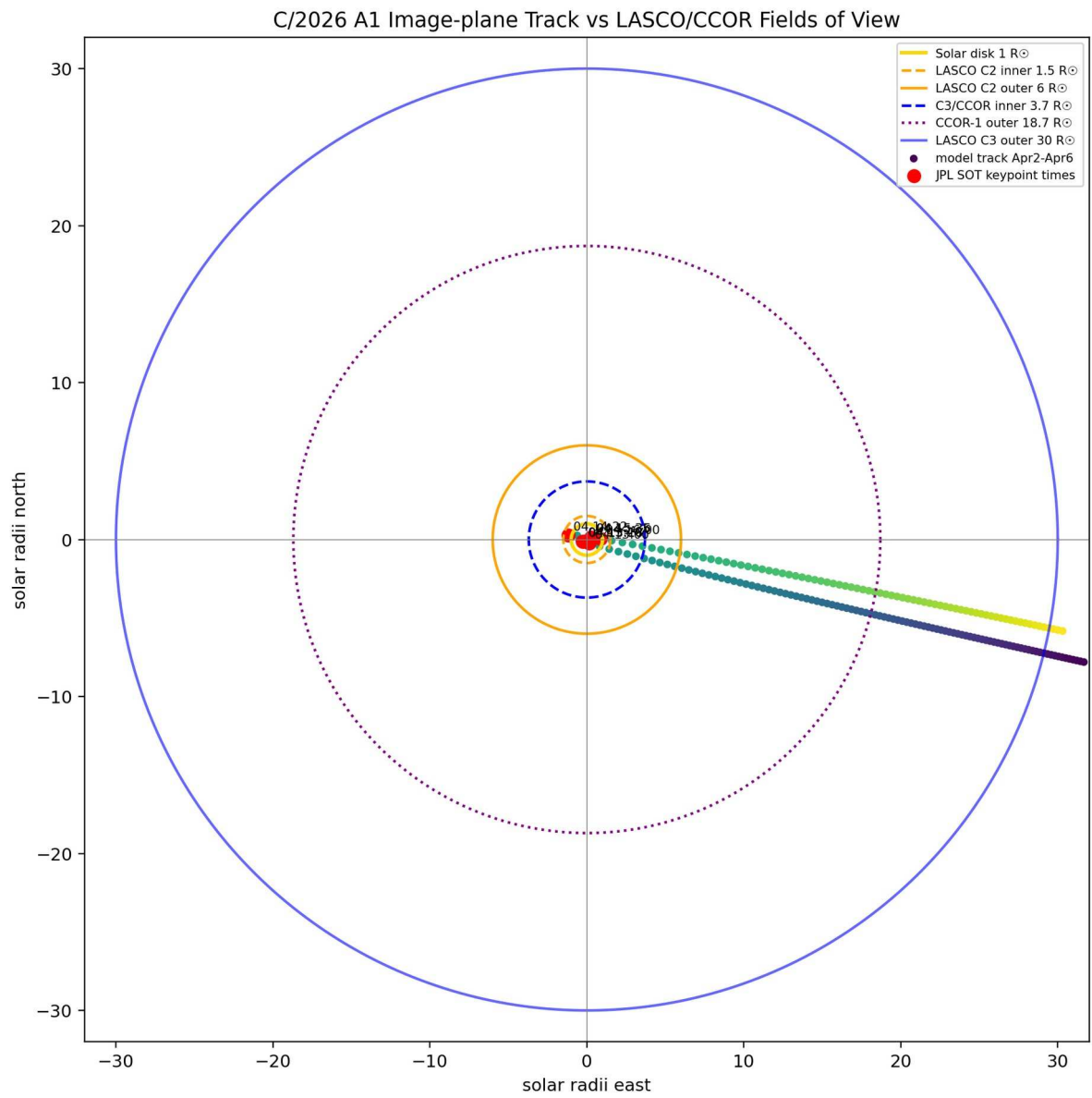


Figure 13. C/2026 A1 image-plane track against LASCO/CCOR fields of view.

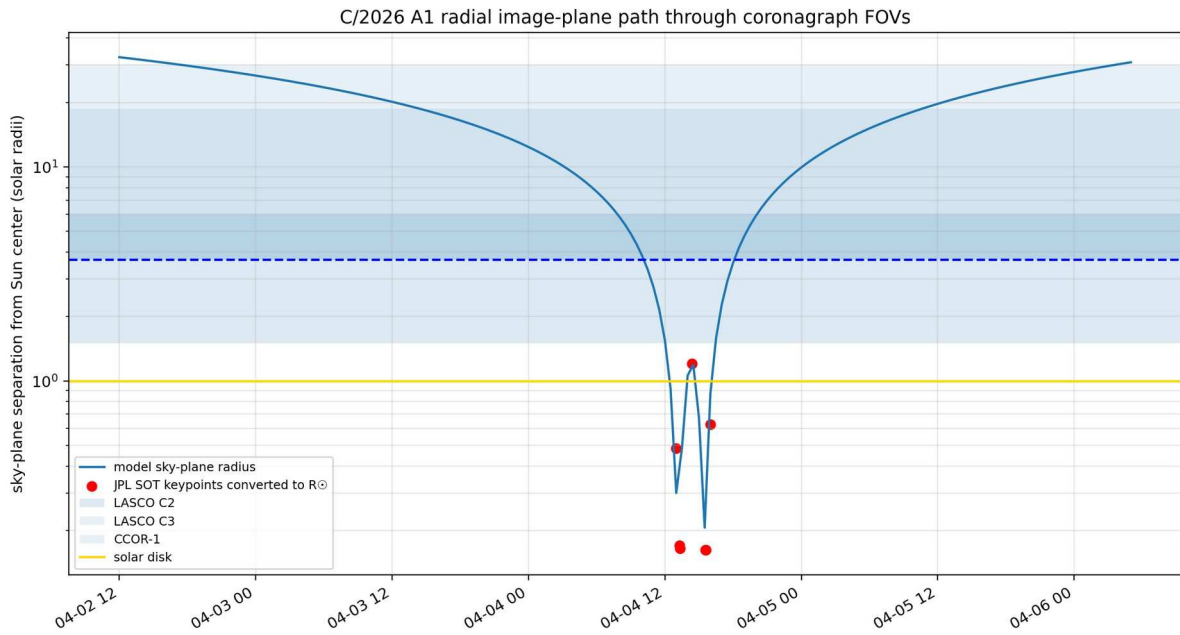


Figure 14. Model sky-plane separation through LASCO/CCOR radial FOVs.

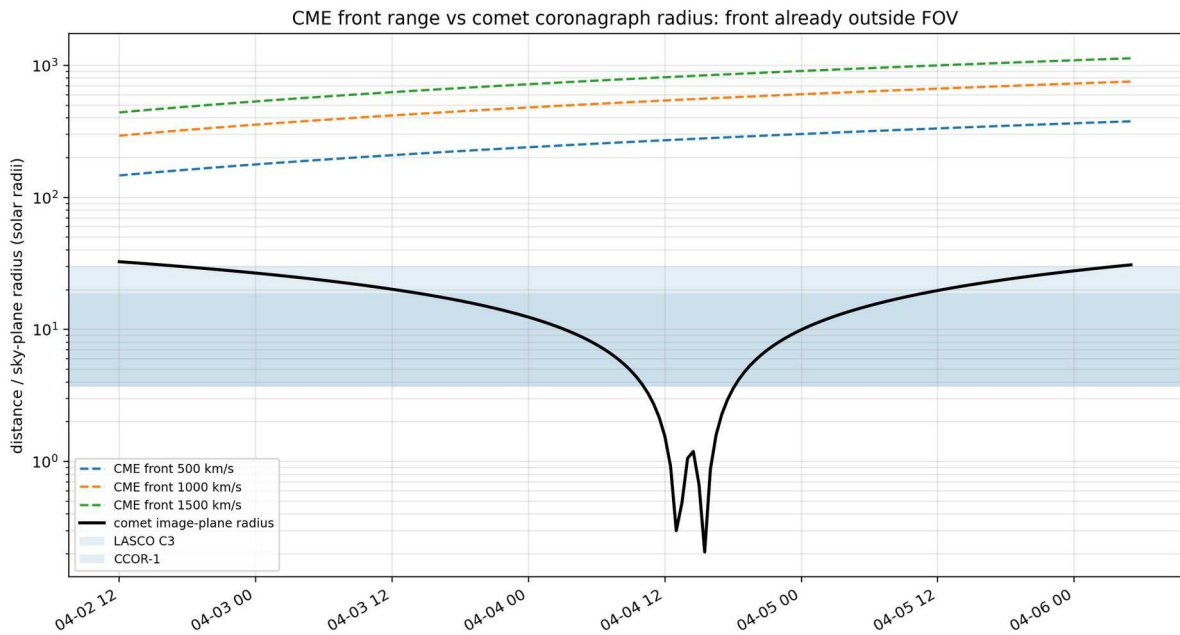


Figure 15. CME front range compared with comet coronagraph radius: the March 31 CME front is already outside the FOV by the Apr 2-6 comet window.

The image-plane result makes the key correction explicit: C/2026 A1 is a coronagraph-frame sungrazer co-event, but a CME front launched on Mar 31 is far beyond the inner coronagraph fields by the time the comet is visible in LASCO/CCOR. A same-frame CME structure would be needed to promote the candidate.

7. Discussion

The research session built a progressively stricter chain of evidence. Early scoring made C/2026 A1 appear to be the best object in the window. Subsequent radial-front modeling and JPL/LASCO/CCOR projection corrected that interpretation. This is the desired scientific behavior: the framework did not merely reinforce a preferred hypothesis; it separated co-event evidence from front-companion evidence.

The strongest affirmative result is therefore not that a CME-carried object was found. The strongest affirmative result is that a structured pipeline can map solar events, orbital objects, coronagraph geometry, and harmonic-LIDAR trackability while retaining negative gates and null-control discipline.

Claim	Current status	Reason
C/2026 A1 was a solar/CME surge co-event	Supported	Temporal and coronagraph-frame alignment around Apr 2-6; near-Sun sungrazer morphology.
C/2026 A1 travelled with the Mar 31 CME front	Not supported	CME front radial distance is far beyond LASCO/CCOR FOV by the comet window.
A station-group object came with a CME	Not supported	No strict candidate; dataset lacks full object catalog and first-seen history.
ISS OBJECT XY / CORAL are drag responders	Watchlist only	High drag response scores, but known station/local family context and no CME provenance.
Harmonic LIDAR can schedule orbital observations	Prototype supported	72-hour scheduler and phase barcode products generated; physical lidar performance remains unmodeled.

8. Falsification and Promotion Rules

A candidate may only be promoted from co-event to CME-linked companion if the following gates pass:

Temporal gate: object detection or morphology change falls inside a defined CME surge window.

Radial gate: object radial distance and CME front radial distance overlap within the same field-of-view interval.

Position-angle gate: object PA(t) overlaps CME-front PA(t) in the same coronagraph frames.

Tracklet gate: independent image-plane centroid sequence can be fitted through the claimed event.

Provenance gate: object is not explained by ordinary launch, debris, known station stack, or normal comet/sungrazer dynamics.

Null gate: equivalent random or control windows do not produce comparable candidate scores.

If any of the radial, PA, or provenance gates fails, the object remains a co-event or control object, not a companion.

9. Next Work

Task	Purpose
LASCO/CCOR frame ingestion	Download or access running-difference frames and extract

	centroid/radius/PA values for the comet and CME structures.
CME front extraction	Build front PA/radius sequences for the Mar 31 CME in the same instrument frame as C/2026 A1.
Catalog-wide orbital search	Run the CME companion scoring on all CelesTrak GP objects plus historical TLE/first-seen records.
SGP4 upgrade	Replace two-body orbital propagation with SGP4 for conjunction and LIDAR scheduling.
Physical lidar model	Add aperture, laser power, pulse repetition frequency, beam divergence, atmospheric transmission, optical cross-section, and detector noise.
External review	Submit a minimal, falsifiable claim package: parser correctness, image-plane co-event map, and explicit non-promotion of the CME-front claim.

References

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<https://ssd.jpl.nasa.gov/api/horizons.api>
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- [10] RA-CONSULTING/aureon-trading. AUREON HNC Surge Detector and Space Weather Bridge source files. <https://github.com/RA-CONSULTING/aureon-trading>

Appendix A: Artifact Inventory

Artifact	Path
LSC mapping proof	/mnt/data/lsc_full_mapping_proof_run_bundle.zip
LSC/HNC requirements run	/mnt/data/lsc_hnc_qgita_requirements_run_bundle.zip
SCEMDA v10 Enterprise Agent	/mnt/data/scemda_v10_enterprise_agent_bundle.zip
Current orbital run	/mnt/data/scemda_current_orbital_run_bundle.zip
Harmonic-LIDAR orbital map	/mnt/data/scemda_harmonic_lidar_orbital_map_bundle.zip
Harmonic-LIDAR 72h scheduler	/mnt/data/ scemda_harmonic_lidar_scheduler_72h_bundle.zip
CME companion candidate screen	/mnt/data/ scemda_cme_companion_candidate_run_bundle.zip
C/2026 A1 JPL/Horizons packet	/mnt/data/c2026a1_jpl_horizons_data_packet_bundle.zip
C/2026 A1 LASCO/CCOR image-plane packet	/mnt/data/c2026a1_lasco_ccor_image_plane_bundle.zip
C/2026 A1 shape/area comparison	/mnt/data/ scemda_object_shape_area_comparison_bundle.zip